Module 02 – Transportation Modeling

Model Formulation

<u>Decision Variables</u>: variables representing the allocation values in the table
<u>Constraints</u>: capacity & SUM
<u>Objective function</u>: total = Sumproduct of all the variables in the table to be changed

Model Optimized for Profit

- *A screenshot of your optimized final model (formatted nicely, of course)*
- *A text explanation of what your model is recommending*

Average of Cost Per	Column Labels								
Row Labels	D0f6eec2	D60bded9	D662583f	D759aba7	D7cefff4	Dc20a1c2	(blank)	Grand Total	Capacity
S43d4c90	0.089999989	0.190000003	0.079999979	0.08000002	0.179999956	0.080000032		0.115181815	145
S5f97721	0.189999993	0.050000005	0.090000038	0.090000013	0.119999998	0.079999951		0.107081338	184
S6e0cec4	0.059999966	0.119999964	0.150000028	0.060000015	0.09000003	0.170000043		0.106404967	134
S7eda83e	0.090000029	0.15000005	0.080000005	0.100000051	0.189999994	0.170000009		0.131681438	157
(blank)									
Grand Total	**0.107935476**	**0.129931514**	**0.100516139**	**0.081007219**	**0.142922075**	**0.126216227**		**0.115083619**	
Demand	109	110	115	122	121	117			

Total Cost	D0f6eec2	D60bded9	D662583f	D759aba7	D7cefff4	Dc20a1c2		SUM	Capacity
S43d4c90	$ -	$ 110.00	$ -	$ -	$ 35.00	$ -		145	145
S5f97721	$ 109.00	$ -	$ 27.00	$ 48.00	$ -	$ -		184	184
S6e0cec4	$ -	$ -	$ 88.00	$ -	$ -	$ 46.00		134	134
S7eda83e	$ -	$ -	$ -	$ -	$ 86.00	$ 71.00		157	157
Demand	109	110	115	122	121	117		Total:	$ 104.09
SUM	109	110	115	48	121	117			

My model is representative of cost analysis fulfilling demand across different supply chain nodes. The top section “average of Cost Per” displays the per-unit cost of supplying goods from various suppliers to different destinations. The “demand” represents the required quantities at each destination and the “capacity” represents the maximum available supply.